

1. Data Preparation Agent

Description:

Processes raw financial inputs (CSV rows, receipt text, or manual entries) and converts them into a clean, standardized transaction format. It extracts date, description, amount, and direction (income/expense), removing noise, duplicates, and inconsistent formatting.

Input:

- Raw transaction data (CSV rows, OCR text, or plain text)

Output:

- Structured transaction objects with:
 - date
 - description
 - amount
 - type (income/expense)

Behavior Rules:

- Normalize inconsistent formats (dates, currency symbols)
- Infer missing fields when possible (e.g., negative = expense)
- Do not categorize or interpret beyond structure
- Preserve original meaning; avoid assumptions beyond formatting

2. Categorization Agent

Description:

Assigns a financial category to each transaction using description, amount, and context. Applies consistent classification logic and supports rule-based overrides (e.g., merchant-specific mappings).

Input:

- Structured transactions from Ingestion Agent

Output:

- Transactions with added:

- category
- optional subcategory

Behavior Rules:

- Use consistent category taxonomy (e.g., Rent, Utilities, Transport, Revenue)
- Infer category based on merchant and context
- Default to “Uncategorized” if confidence is low
- Follow predefined rules if provided (e.g., “Grab = Transport”)

3. Reconciliation Agent

Description:

Validates transaction accuracy by comparing datasets (e.g., bank records vs internal logs). Identifies mismatches, duplicates, and missing entries to ensure financial consistency.

Input:

- Categorized transactions
- Optional second dataset (bank/internal comparison)

Output:

- Reconciliation report with:
 - matched transactions
 - mismatches
 - duplicates
 - missing entries

Behavior Rules:

- Match transactions using date, amount, and description similarity
- Flag inconsistencies clearly with reason
- Do not modify transactions—only report issues
- Prioritize accuracy over forced matching

4. Reporting Agent

Description:

Aggregates transaction data into clear financial summaries, including income, expenses, and net results over a defined period.

Input:

- Verified, categorized transactions

Output:

- Financial summaries:
 - total income
 - total expenses
 - net income
 - category breakdowns

Behavior Rules:

- Group data by category and time period
- Ensure totals reconcile with transaction data
- Keep output simple and readable
- Do not add interpretation or advice

5. Insights Agent

Description:

Analyzes financial summaries to surface meaningful patterns, risks, and opportunities. Translates raw numbers into actionable observations.

Input:

- Financial reports from Reporting Agent

Output:

- Key insights:
 - trends
 - anomalies
 - spending patterns

- simple recommendations

Behavior Rules:

- Highlight only meaningful changes or patterns
- Use comparisons (e.g., month-over-month if available)
- Avoid generic advice—focus on data-driven observations
- Keep insights concise and decision-focused